Reinforcement learning from human feedback

Reward models, the KL penalty and PPO

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ETH Zurich

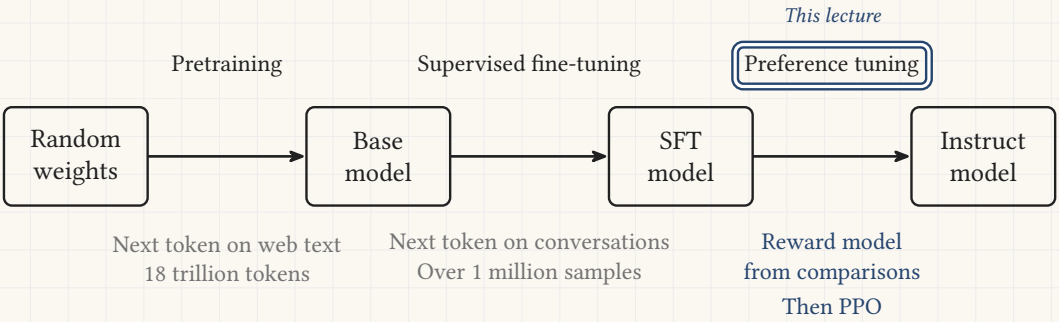
20 October 2026

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Section 1

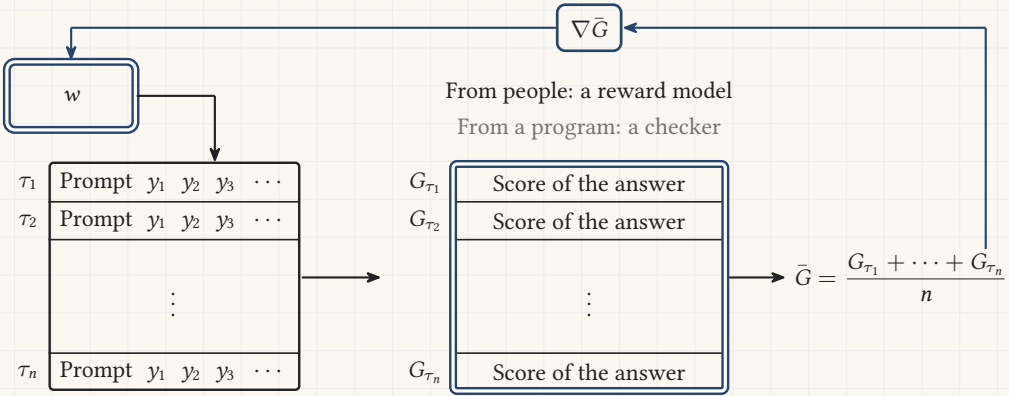
Learning from comparisons

One more training stage



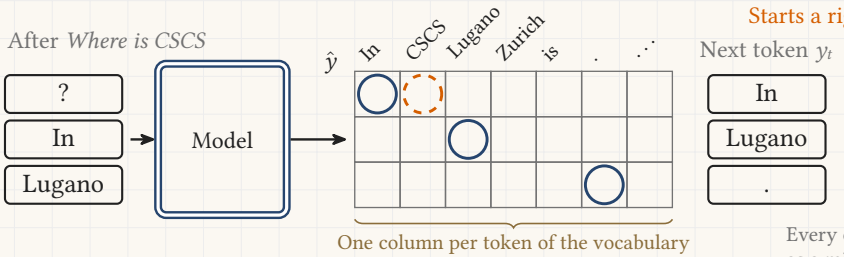
Qwen2.5 Technical Report, arXiv:2412.15115

The same loop for a language model



The SFT loss reads one answer

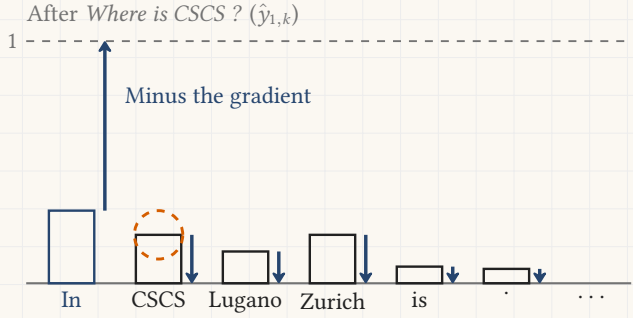
SFT loss $\longrightarrow \bar{l} = -\frac{1}{3}\underbrace{\left(\log \hat{y}_{1,\text{In}} + \log \hat{y}_{2,\text{Lugano}} + \log \hat{y}_{3,.}\right)}_{\text{Circled probabilities}}$



A gradient step may lower other correct tokens

Row 1's cross-entropy → $\dfrac{\partial\,(-\log \hat{y}_{1,\text{In}})}{\partial z_{1,k}} = \begin{cases} \hat{y}_{1,k} - 1 & k = \text{In} \\ \hat{y}_{1,k} & \text{every other } k \end{cases}$

Logit of token k →



A right answer

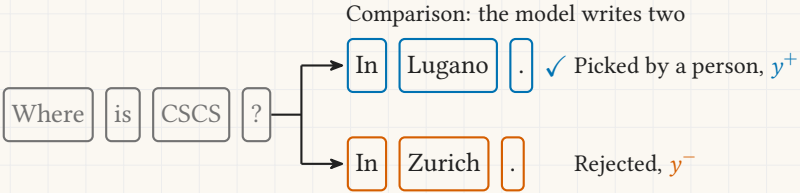
CSCS | is | in | Lugano | .

A wrong answer

Zurich | .

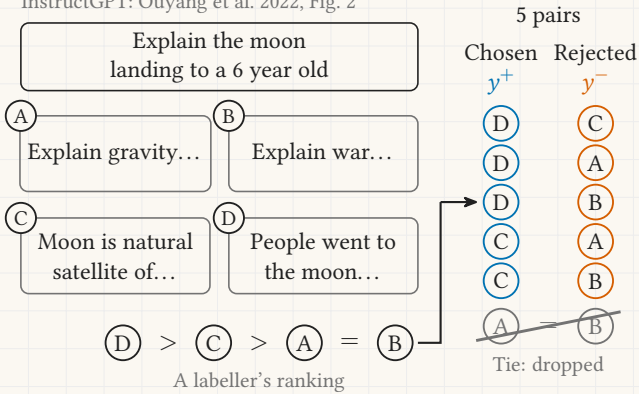
The step lowers CSCS exactly as it lowers Zurich.

A comparison in place of a demonstration



Rankings become pairs

InstructGPT: Ouyang et al. 2022, Fig. 2



4 to 9 answers per prompt — Sec. 3.5

$\binom{4}{2} = 6$ to $\binom{9}{2} = 36$ pairs

About 40 labellers — Sec. 3.4

33,207 ranked prompts — Table 6

Labeller agreement — Sec. 3.4

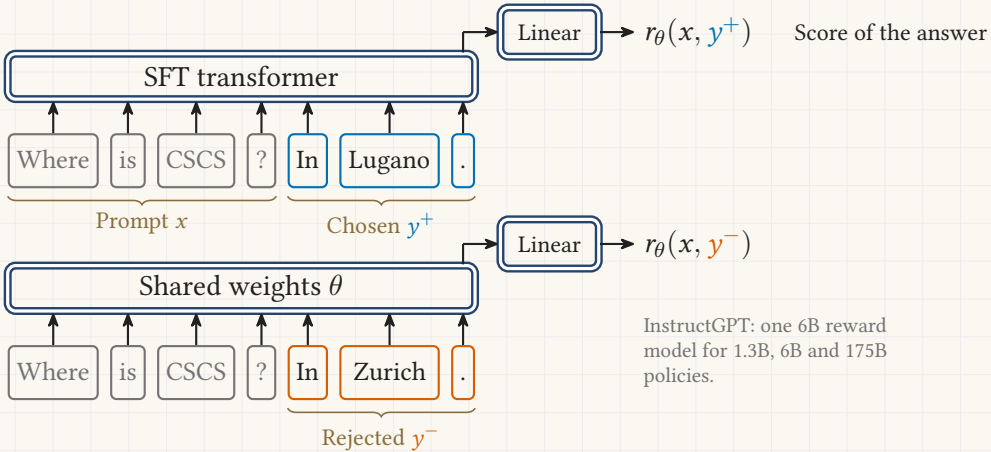
$72.6 \pm 1.5\%$

2

Section 2

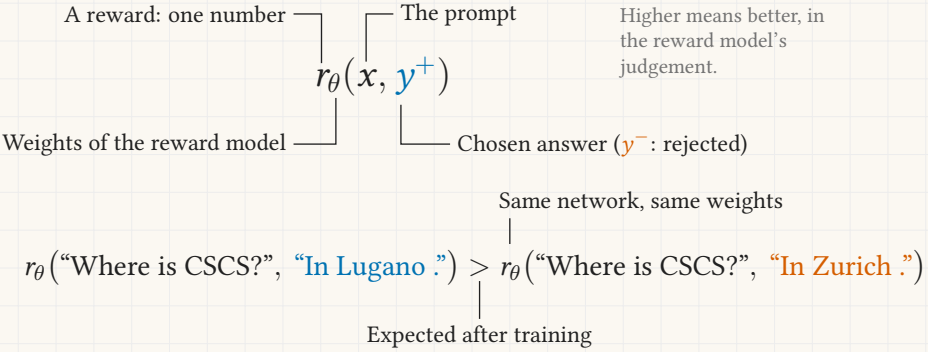
The reward model

The reward model



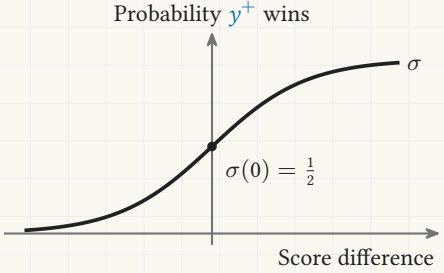
InstructGPT: one 6B reward model for 1.3B, 6B and 175B policies.

The notation $r_\theta(x, y^+)$



The Bradley-Terry model

$$P\big(\underbrace{y^+ \succ y^-}_{\text{Chosen over rejected}} \mid x\big) = \frac{e^{r_\theta(x,y^+)}}{e^{r_\theta(x,y^+)} + e^{r_\theta(x,y^-)}} = \sigma\big(\underbrace{r_\theta(x, y^+) - r_\theta(x, y^-)}_{\text{Score difference}}\big)$$

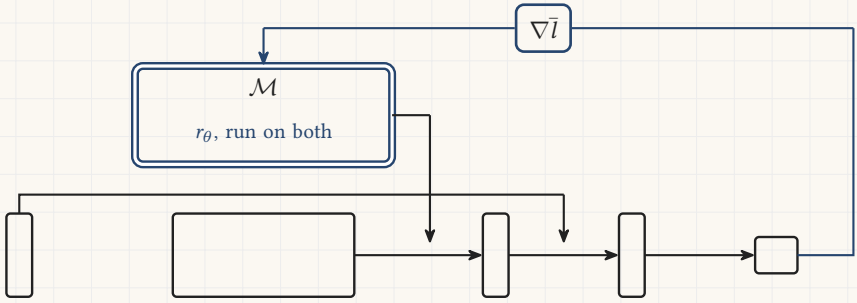


Bradley and Terry 1952
Christiano et al. 2017, eq. (1)

The model behind Elo ratings in chess.

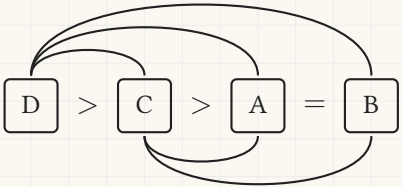
Training the reward model

$$\hat{y} = \sigma\big(r_\theta(x, y^+) - r_\theta(x, y^-)\big) \qquad l = -\log \hat{y}$$



The loss over one ranking

$$\text{loss}(\theta) = -\operatorname*{avg}_{\text{prompts } x}\left[\operatorname*{avg}_{\text{pairs of } x}\underbrace{\log\sigma\big(r_\theta(x, y^+) - r_\theta(x, y^-)\big)}_{\text{One pair}}\right]$$



5 pairs

X	c
(x, D, C)	1
(x, D, A)	1
(x, D, B)	1
(x, C, A)	1
(x, C, B)	1

One example per pair

Winner listed first: $c = 1$

All pairs of one prompt in one batch element; one epoch.

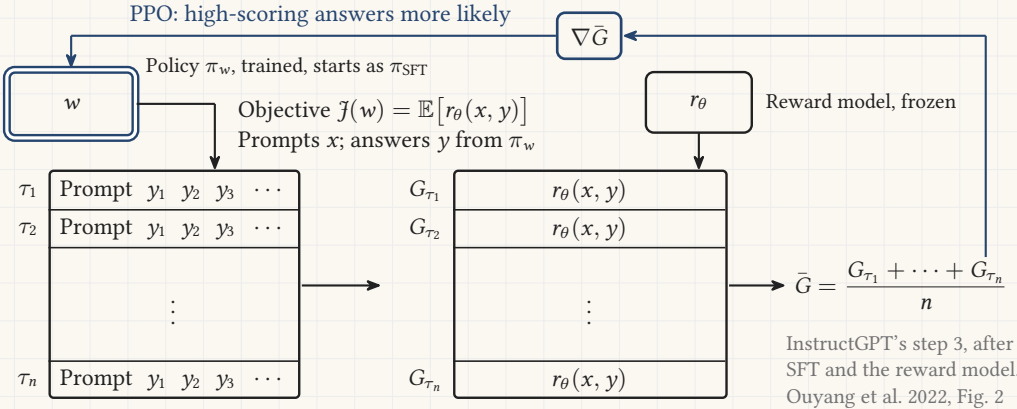
Ouyang et al. 2022, eq. (1), Fig. 2

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Section 3

The KL-regularised objective

Reinforcement learning against the reward model

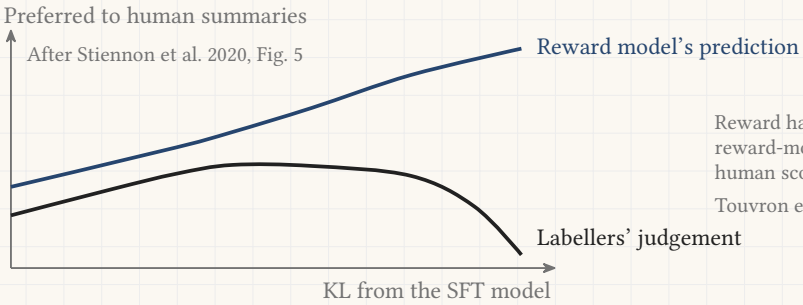


InstructGPT's step 3, after SFT and the reward model. Ouyang et al. 2022, Fig. 2

Maximising the reward model's score over-optimises

$$\underbrace{\mathcal{J}(w)}_{\text{Objective}} = \mathbb{E}\Big[\underbrace{r_\theta(x, y)}_{\text{Reward model's score}}\Big]$$

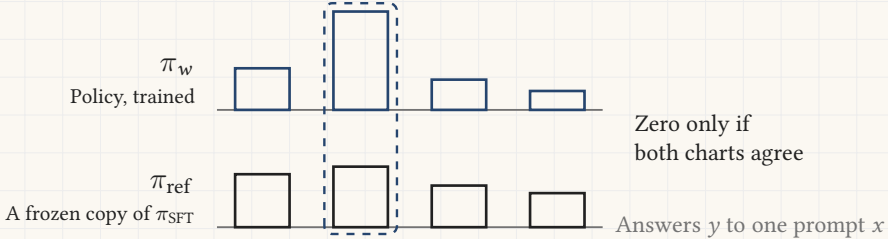
Prompts x; answers y from the policy π_w



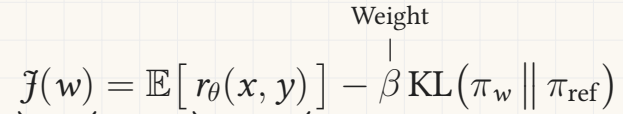
The Kullback-Leibler divergence

$$\mathrm{KL}\left(\pi_w \,\middle\|\, \pi_{\mathrm{ref}}\right) = \mathbb{E}\left[\underbrace{\log \frac{\pi_w(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)}}_{\text{Log-ratio of one answer } y}\right] \geq 0$$

Over answers y drawn from π_w



The KL-regularised objective



Prompts x; answers y from the policy π_w

Reference π_{ref}: a frozen copy of π_{SFT}

Keeps y where r_θ was trained

Stiennon et al. 2020

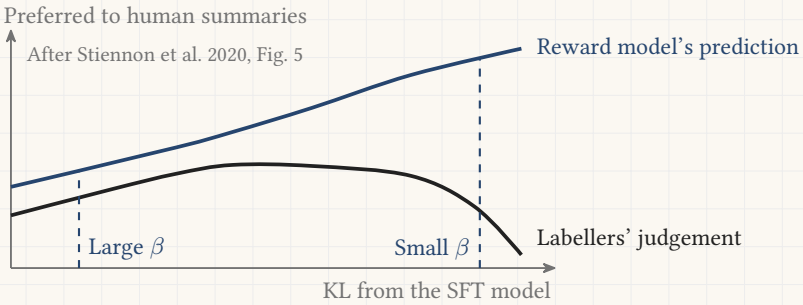
Keeps the answers varied

$\beta = 0.02$	$\beta = 0.05$	$\beta = 0.01$	$\beta = 0.005$
InstructGPT	Stiennon et al. 2020	Llama 2, 7B and 13B	Llama 2, 34B and 70B

The weight β trades reward against drift

InstructGPT: best β near 0.01 to 0.02;
$\beta = 0$ and $\beta = 2$ both poor
Ouyang et al. 2022, App. E.7

Penalty acts like early stopping
Gao et al. 2023

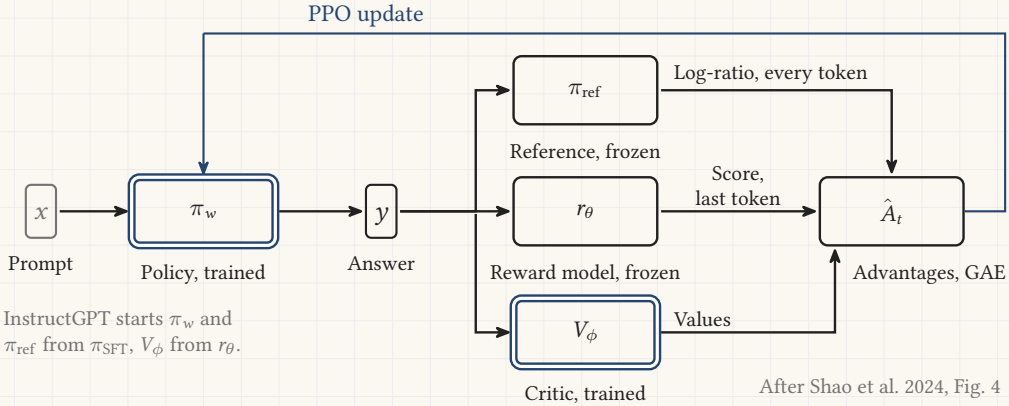


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Section 4

PPO against a reward model

Four networks



InstructGPT starts π_w and π_{ref} from π_{SFT}, V_ϕ from r_θ.

After Shao et al. 2024, Fig. 4

The alignment tax

$$\mathcal{J}(w) = \underbrace{\mathbb{E}\big[\, r_\theta(x, y) \,\big] - \beta\, \mathrm{KL}\big(\pi_w \,\big\|\, \pi_{\text{ref}}\big)}_{\text{KL-regularised objective}} + \underbrace{\eta\, \mathbb{E}\big[\, \log \pi_w(\text{text}) \,\big]}_{\text{Pretraining data, weight } \eta}$$

PPO

SQuAD | DROP | HellaSwag | WMT, French to English

Below GPT-3

InstructGPT: $\eta = 27.8$

Ouyang et al. 2022, eq. (2), where it is γ

Regressions greatly reduced

Labeller preference kept

References

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Notation

x, y	A prompt, an answer	y^+, y^-	Chosen, rejected answer
y_t	Token t of y	c	Labeller's choice, 1 if y^+ won
π_w	The policy, trained; weights w	$\sigma(z)$	$1/(1+e^{-z})$
π_{SFT}	The model after SFT	β	Weight of the KL penalty
π_{ref}	A frozen copy of π_{SFT}	η	Weight of the pretraining term
r_θ	The reward model, weights θ	V_ϕ, $\hat{A}_t$	Critic, advantage estimate
$\mathbb{E}[\,\cdot\,]$	Expectation	$\mathrm{KL}(p \,\|\, q)$	Kullback-Leibler divergence